

TABLE V: RoboTwin 2.0 Benchmark Full Results.

Model	Adjust Bottle Moving PlayingCard away	Beat Hammer Block Lift Pot	Click Bell Place Burger Fries	Dump Bin BigBin Place Can Basket	Avg
π_0	97% 79%	78% 92%	35% 85%	89% 64%	77.38%
π_0 w/ object head	99% 91%	94% 95%	62% 93%	94% 66%	<u>86.75%</u>
π_0 w/ skill head	<u>98%</u> 93%	92% <u>96%</u>	39% <u>95%</u>	94% 73%	85.00%
π_0 w/ depth head	<u>98%</u> <u>97%</u>	96% 82%	<u>63%</u> 94%	87% <u>74%</u>	86.38%
π_0 w/ all heads (Ours)	99% 98%	<u>95%</u> 99%	65% 98%	<u>93%</u> 78%	90.63%

the depth features. We introduce a control parameter $\delta \in [0, 1]$ (referred to as the "depth feature ratio" in the analysis) to construct the ablated feature representation $\tilde{\mathbf{f}}$:

$$\tilde{\mathbf{f}} = \delta \cdot \mathbf{f}_{\text{depth}} + (1 - \delta) \cdot \epsilon. \quad (12)$$

These corrupted features $\tilde{\mathbf{f}}$ are then projected into keys K_{Depth} and values V_{Depth} for the specific attention heads. By varying δ from 0 (pure noise) to 1.0 (clean depth signal), we quantitatively evaluate how the quality of 3D structural information impacts manipulation success.

Skill Head. To examine the causal effect of skill recognition on task success, we regulate the model's intent classification accuracy to specific target levels $\gamma \in \{0.25, 0.5, 0.75, 1.0\}$. We define the *soft accuracy* S as the mean predicted probability assigned to the ground-truth skill class y_i across a batch of size N :

$$S = \frac{1}{N} \sum_{i=1}^N \hat{p}_i(y_i), \quad (13)$$

where $\hat{p}_i(y_i)$ denotes the probability of the correct label derived from the softmax distribution. To enforce convergence to the target accuracy γ , we introduce an auxiliary control loss $\mathcal{L}_{\text{ctrl}}$ derived from the Smooth L1 distance:

$$\mathcal{L}_{\text{ctrl}} = \begin{cases} \frac{0.5(S - \gamma)^2}{\beta}, & \text{if } |S - \gamma| < \beta, \\ |S - \gamma| - 0.5\beta, & \text{otherwise.} \end{cases} \quad (14)$$

with β set to 0.02. The final objective for the skill head during ablation is a weighted sum: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{skill}} + \lambda \mathcal{L}_{\text{ctrl}}$. By adjusting the weight λ , we empirically ensure the model's skill recognition performance converges to the designated target.

To measure the intrinsic skill representation capability of the baseline π_0 (reported as 48.4%), we employ the identical skill head architecture and classification objective ($\mathcal{L}_{\text{skill}}$) as described above. Specifically, we attach the projection layer ($\mathbf{f} \rightarrow \hat{\mathbf{p}}$) to the pre-trained π_0 . Distinct from the controlled ablation models, we freeze the entire backbone and exclusively

optimize the projection parameters ($\mathbf{W}, \mathbf{b}$) using the classification loss. This setup effectively functions as a linear probe on the fixed action features to evaluate their linear separability regarding skill semantics.

Considering the LIBERO dataset consists of 3 task-level skill categories, a purely random guess over the task-level skills would yield an accuracy of $\sim 33.3\%$. In implementation, we use four classifier outputs: the three effective skills plus one null/background class for unannotated or transition frames. Consequently, the baseline's performance of 48.4% represents only a marginal improvement over chance. This indicates that without explicit temporal logic supervision, the representation of π_0 captures negligible high-level intent information, failing to effectively disentangle the long-horizon structure of tasks.

E. Ablation on Head and Adapter Design Choices

This part provides detailed explanations corresponding to Section V-E of the main paper.

1) *Object Head:* To improve the interpretability and spatial alignment of attention in action decoding, we supervise a dedicated set of heads $\mathcal{H}_{\text{obj}}$ to focus on semantically meaningful regions—such as the object to be grasped or its intended destination. We investigate two strategies for supervising these attention heads: a binary mask-based region loss, and a Gaussian prior-based KL divergence loss.

Object Region Supervision. To guide a subset of heads $\mathcal{H}_{\text{obj}}$ toward attending semantically meaningful areas such as the grasp object or destination region, we use direct supervision from binary masks $\mathbf{M}$ annotated by foundation models. These masks indicate which patches are considered object-relevant. Consistent with Eq. 4, the supervision loss is defined as a negative log likelihood over the total attention mass inside the valid object region:

$$\mathcal{L}_{\text{object}} = -\frac{1}{|\mathcal{T}_a|} \sum_{t \in \mathcal{T}_a} \log \left(\max_p \left(\sum_p \bar{P}_{t,p} M_p, \epsilon \right) \right). \quad (15)$$

Importantly, $\mathbf{M}$ only specifies which patches are object regions while leaving the distribution inside those regions